

Priyanshu Dhingra

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🌐 Website 📄 Google Scholar 🐙 GitHub 🔗 LinkedIn

Objective

MScAC AI student and former AI Software Engineer with production experience building LLM agents, AI systems, and evaluation pipelines, alongside ML research experience in computer vision and speech. Interested in applying and scaling ML research into reliable real-world AI systems.

Experience

Teciem (formerly Finastra - Treasury & Capital Markets BU)

January 2025 – July 2026

AI Software Engineer | 🌀 Project: Kondor AI Assistant

- Improved NL2SQL reliability and reduced per-query token usage by **30%** by engineering a **dependency schema AI agent** using LangGraph, LangChain, and Azure OpenAI to model optimized SQL join paths.
- Designed a deterministic **event-hyperlink generation** pipeline by extracting deal identifiers and validating eligibility through SQL checks, replacing an LLM-based approach to ensure consistent and accurate hyperlink generation.
- Enforced **access controls** on AI-generated SQL for GDPR/SOC2 compliance, ensuring **100%** data security.
- Reduced manual annotation effort by **95%** by developing an **LLM-driven automation framework** that transformed over **1,000** unstructured QA business cases into JSON-based evaluation artifacts.
- Improved LLM response **accuracy to 96%** through systematic prompt engineering, **RAG optimization**, and evaluation framework design.
- Reduced downtime by over **50%** and improved operational reliability by diagnosing and resolving **20+ critical bugs**.
- Improved software reliability by implementing automated **unit and integration tests**, increasing code coverage and reducing regression issues.

Carnegie Mellon University

September 2024 – January 2025

Computer Vision Research Intern | 🌀 Project: Vision transformer for 3D Subtomogram Alignment

- Co-authored a paper accepted at **IEEE/CVF CVPR 2025** by contributing to literature review, model design, experiments, and developing a reproducible **PyTorch codebase**.
- Achieved **77% lower rotation error** and **62% lower translation error** at low SNR by contributing to the design and implementation of **Polyshift** and **MARE** that improved shift/rotation equivariance in ViT-based 3D alignment.
- Built synthetic **3D subtomogram** generation and **2D-slice visualization** pipelines to support dataset construction and experimental evaluation.
- Implemented a **Fourier-domain missing-wedge mask** and interpolation pipeline to improve alignment robustness and reduce anisotropic artifacts.
- Debugged and optimized BOE-ViT training to enable **large-batch scalability** and reproducible experiments.

Nanyang Technological University

March 2024 – August 2024

Speech & Language AI Intern | 🌀 Project: Automatic Speech De-identification Systems

- Published **first-author research** in **three IEEE conferences (IALP 2024, TENCON 2024, ICAICTA 2024)** and **one journal (IJALP 2024)** on speech privacy and de-identification.
- Achieved **0.79 F1** for detecting and masking **Personally Identifiable Information (PII)** in audio, implementing anonymization pipelines aligned with privacy-preserving ML principles.
- Addressed PII data scarcity by innovating an **LLM-based synthetic data generation pipeline**, paired with **WhisperSpeech TTS** to create speech-text pairs.
- Outperformed pipeline models by **6.4% in PII detection accuracy** by designing and validating an **Entity-Aware ASR** model with joint ASR-NER training.

Education

University of Toronto

September 2026 – Present

Master of Science in Applied Computing (AI Concentration)

Coursework (Ongoing): *Neural Networks and Deep Learning, Imitation Learning for Robotics, Probabilistic Learning and Reasoning (Audit), Natural Language Computing, Empirical Research Methods in Software Engineering.*

Rajiv Gandhi Institute of Petroleum Technology, Rae Bareli

December 2021 – May 2025

Bachelor of Technology in Information Technology

CPI: 9.36/10.00

Coursework: *Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management.*

Skills

Languages Python, SQL, C++

Agentic Systems LLM Agents, Multi-agent orchestration, LangGraph, LangChain, RAG, Vector stores, ChromaDB

ML & Modeling PyTorch, Transformers, Self-attention, Scikit-Learn, Training, Evaluation, Inference

Backend & APIs FastAPI, Flask, REST, OpenAPI, Swagger, Postman

Production & MLOps Docker, Git, GitHub, Pytest, Monitoring & Debugging, Deployment Automation

Soft Skills Technical Communication, Analytical Thinking, Adaptability, Problem Solving, Agile Teamwork